

HHS3190M
HUMAN PHYSIOLOGY AND FUNCTIONAL ANATOMY
FOR REHABILITATION SERVICES
HHS3190MJ

復康服務之人體生理學及功能性解剖學

Physiology L1: Human Body and Chemicals
生理學 L1: 人體與化學

(Sem 1 AY26-27 26-27學年第1學期)

Outline大綱

- Background: What is physiology?
背景：什麼是生理學？
- Organization of human body
人體的組織結構
- Overview of organ systems in human body
人體器官系統概述
- Important chemicals in human body
人體重要化學物質

Background: What is physiology?

背景：什麼是生理學？

What is “physiology”? 什麼是「生理學」？

- Physiology is scientific study of: 生理學是關於以下方面的科學研究：
 - **normal conditions and activities of a living organism**
生物的正常狀態和活動
 - **and functions and mechanisms of body parts for maintaining survival.**
及維持生存所需的身體不同部位的功能和機制。
- Content of “**human physiology**”: 「人體生理學」內容：
 - **(1) Normal conditions of human body** 人體的正常狀態
 - **(2) Functions of human cells, tissues, and organs** for achieving normal conditions 達到人體正常狀態所需的細胞、組織和器官的功能

Background: What is physiology?

背景：什麼是生理學？

- The structures and positions of human body parts are highly related to their functions!

人體部位的結構和位置與其功能高度相關！

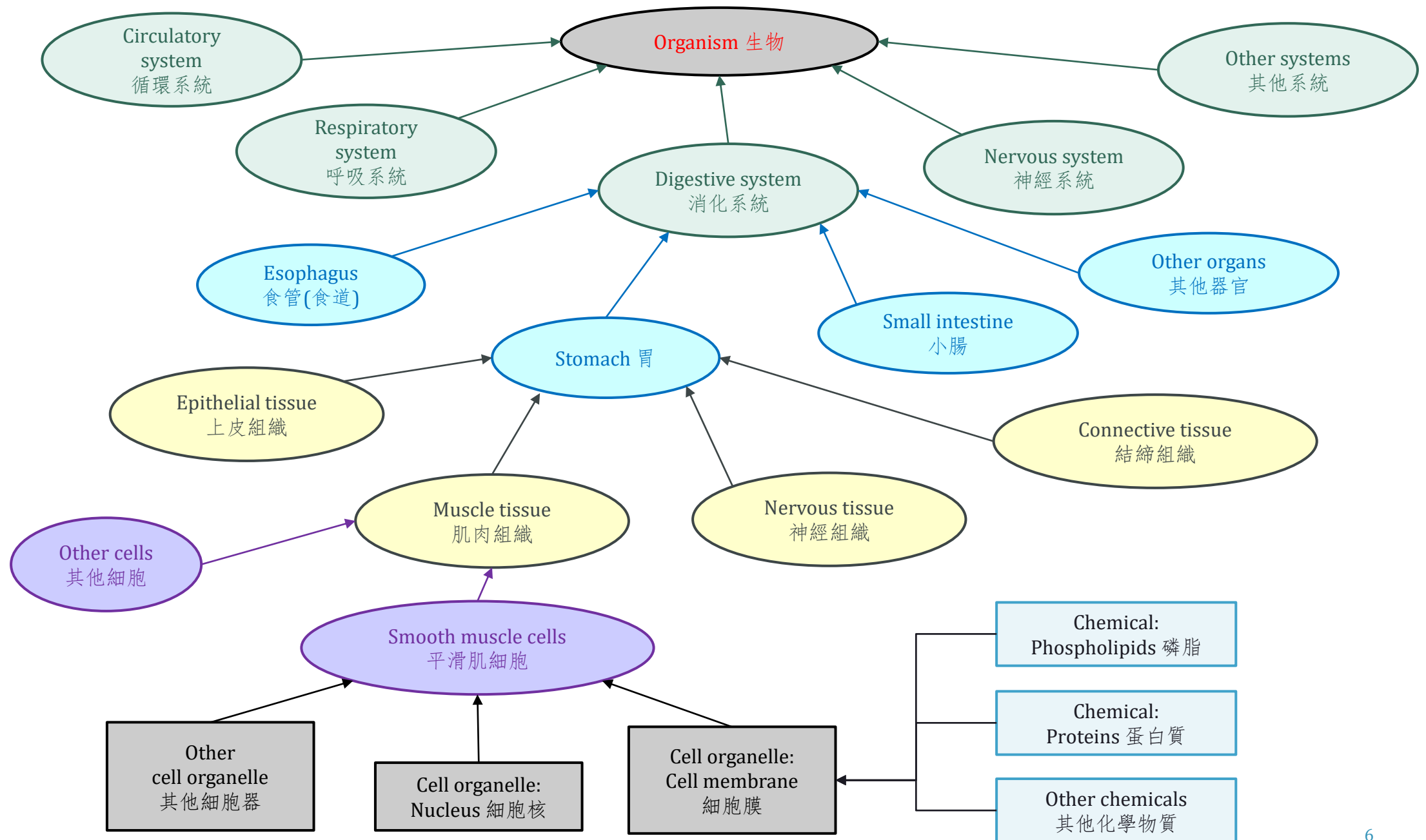
- Thus, learning **(3) basic anatomy of each organ** is essential for understanding the physiological functions!

因此，學習**(3)每個器官的基本解剖結構**對於理解生理功能至關重要！

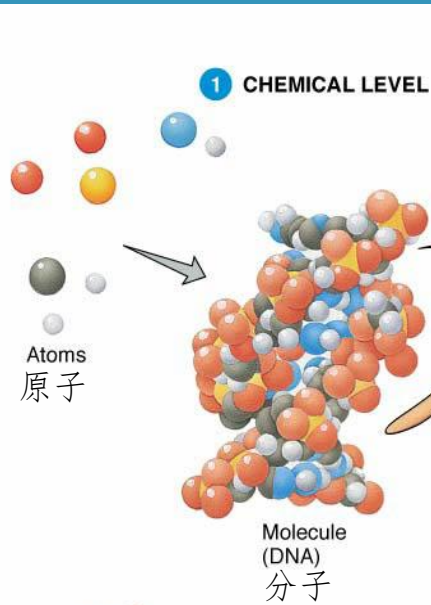
Organization of Human Body

人體的組織結構

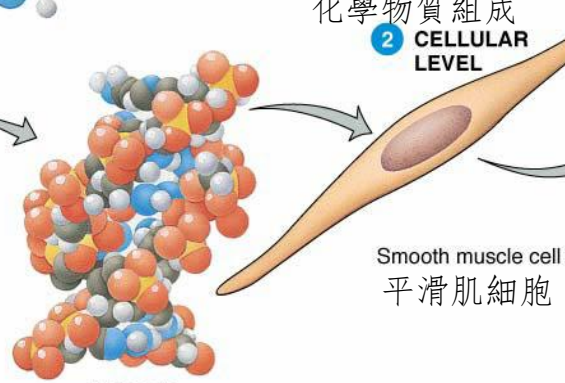
- The organization of the human body can be broken down into several levels, from the simplest to the most complex, namely:
人體的組織結構可以分解成多個層次，從最簡單到最複雜，分別是：
- **Chemical level:** different chemicals (E.g. proteins)
化學層面：不同的化學物質(例如蛋白質)
- **Cellular level:** single cell
細胞層面：單一細胞
- **Tissue level:** groups of similar cells
組織層面：由相似的細胞形成的細胞群體
- **Organ level:** discrete structure, which contains two or more types of tissues, for specific functions
器官層面：由兩個或以上的組織組成的獨立結構，用來執行特定的功能
- **Organ system level:** several organs that work closely together
系統層面：緊密協作的多個器官
- **Organism level:** all organ systems combined to make the whole organism
生物體層面：所有器官系統共同構成整個生物體



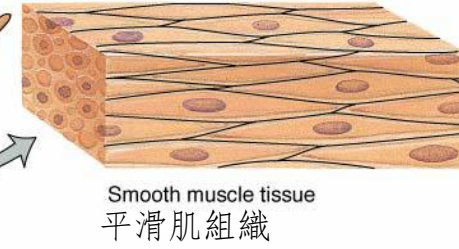
Atoms are combined to form different molecules
原子結合形成不同的分子



Each cell is made up of different chemicals
每個細胞由不同的化學物質組成

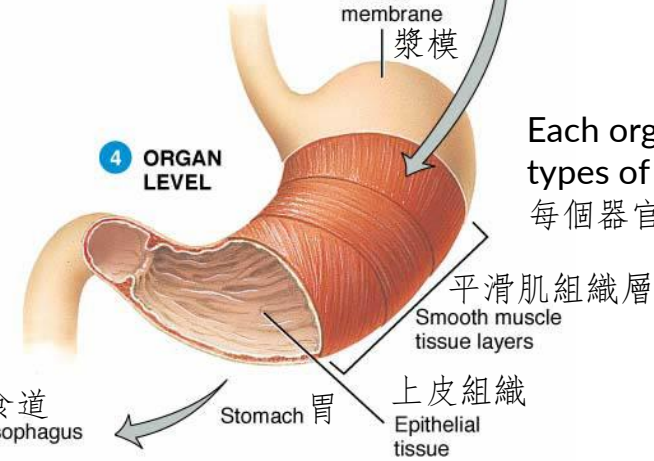


3 TISSUE LEVEL



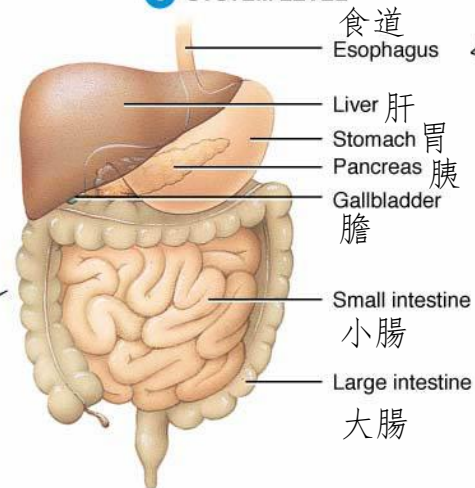
Tissue consists of similar types of cells
組織由相似類型的細胞組成

4 ORGAN LEVEL



Each organ is made up of types of tissues
每個器官都由各種組織組成

5 SYSTEM LEVEL



Each organ system consists of several organs that work together
每個系統都由幾個協同工作的器官組成

The human body is made up of many organ systems
人體由許多系統組成



6 ORGANISMAL LEVEL

Digestive system 消化系統

Important Chemicals in Human Body

人體重要化學物質

- Human body is made of chemicals 人體是由化學物質構成的
- Four types of important chemicals in human body 人體內的四類重要化學物質
 - Water 水
 - Carbohydrates 碳水化合物
 - Lipids 脂肪
 - Proteins 蛋白質

Important Chemicals in Human Body

人體重要化學物質

- Water 水
 - Most abundant inorganic compound (60%–80% of volume of living cells)
最豐富的無機化合物(佔活細胞的體積60%–80%)
 - Important functions: 重要功能:
 - (1) **Useful solvent** 用作溶劑
 - Dissolves many substances for chemical reactions in body
溶解許多用於體內化學反應的物質
 - Body's major transport medium 身體的主要運輸介質
 - (2) **Stabilize body temperature** 穩定體溫
 - Example: Sweating 例如：出汗
 - (3) **Cushioning to protect organs** 保護器官的緩衝
 - Example: cerebrospinal fluid cushions nervous system organs
例如：腦脊髓液為神經系統器官提供緩衝
 - (4) **Lubricant to enhance movement** 用作潤滑劑協助活動
 - Example: synovial fluid in joints 例如：關節滑液

Important Chemicals in Human Body

人體重要化學物質

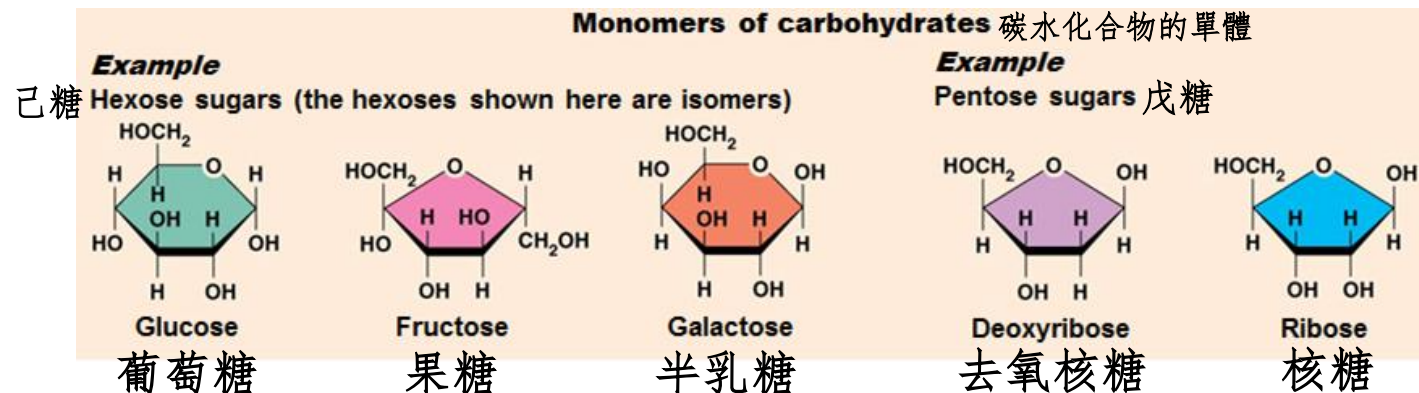
- Carbohydrates 碳水化合物
 - Simple sugars and complex carbohydrates
單糖和複合碳水化合物
 - Molecules are made up of 3 types of atoms - C, H, and O
分子由3種原子組成 - C、H 和 O
 - Three classes 3類
 - (1) **Monosaccharides**: simple sugars
單糖：簡單的糖
 - (2) **Disaccharides**: combination of 2 monosaccharides
雙糖：兩個單糖的組合
 - (3) **Polysaccharides**: complex carbohydrates
多糖：複合碳水化合物

Important Chemicals in Human Body

人體重要化學物質

- **Monosaccharides 單糖**

- Contain 3-7 carbon atoms in each molecules 每個分子含有 3-7 個碳原子
- General formula: $(CH_2O)_n$ (n: number of carbon atoms)
化學通式: $(CH_2O)_n$ (n: 碳原子數)
- Common monosaccharides 常見單糖
- **Glucose** (blood sugar): **葡萄糖**(血糖):
- Directly used by cells for **energy production** 直接被細胞用於生產能量



Important Chemicals in Human Body

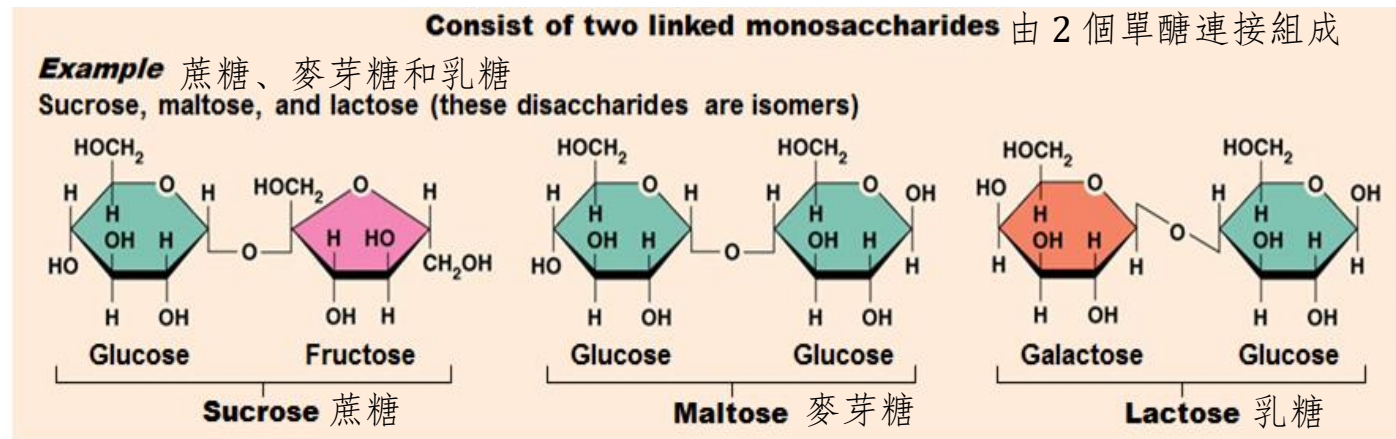
人體重要化學物質

- **Disaccharides** (Double sugars) 二糖(雙糖)
- Combination of two monosaccharides by dehydration
兩個單糖以脫水反應結合
- Too large to pass through cell membranes
太大而無法穿過細胞膜
- Must be broken into monosaccharides before entering cells
必須分解成單糖才能進入細胞

Important Chemicals in Human Body

人體重要化學物質

- Common disaccharides 常見二糖
- **Sucrose** (table sugar): Glucose + Fructose
蔗糖 (食糖): 葡萄糖 + 果糖
- **Maltose**: Glucose + Glucose
麥芽糖: 葡萄糖 + 葡萄糖
- **Lactose** (milk sugar): Glucose + Galactose
乳糖: 葡萄糖 + 半乳糖

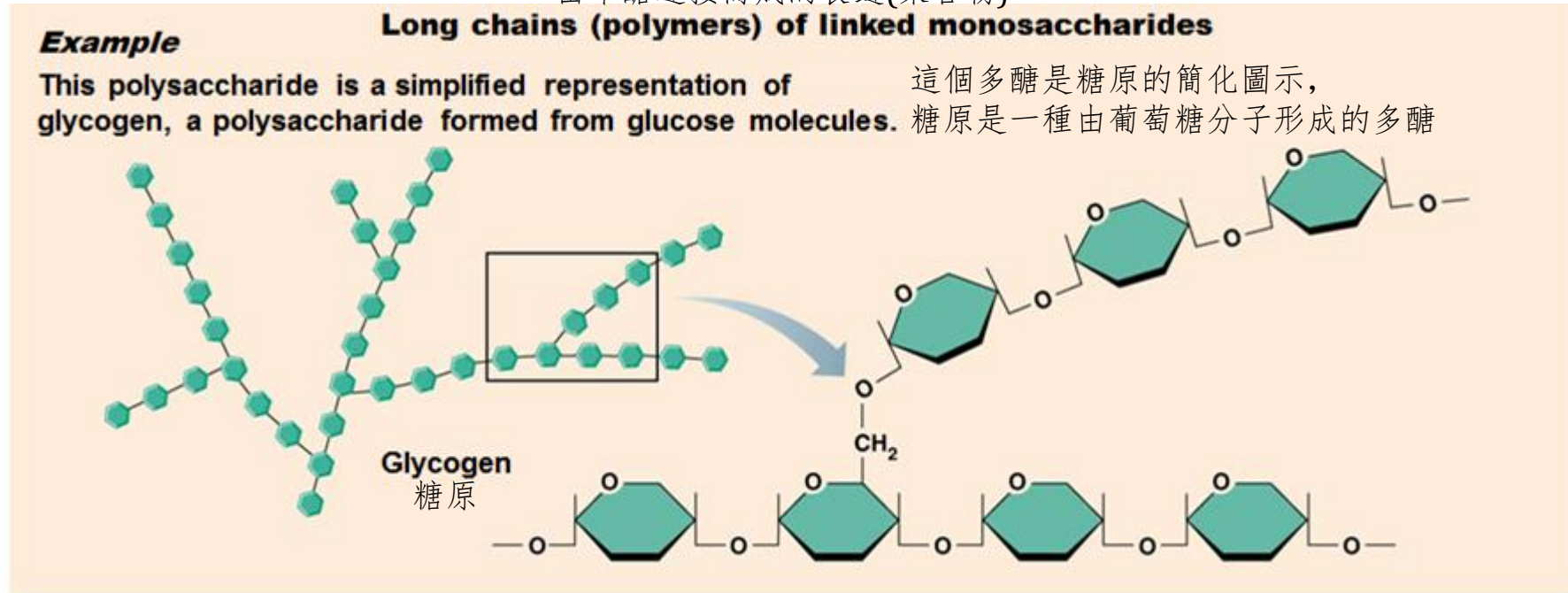


Important Chemicals in Human Body

人體重要化學物質

- **Polysaccharides** (Complex carbohydrates) **多醣** (複合碳水化合物)
 - Formed by combination of many monosaccharides 由多個單醣組合而成
- Common polysaccharides 常見多醣
 - **Fiber** (Cellulose): Polysaccharide that forms cell wall of plant cell
纖維 (纖維素): 形成植物細胞之細胞壁的多醣
 - **Starch**: carbohydrate storage form used by plants
澱粉: 植物使用的碳水化合物儲存形式
 - **Glycogen**: carbohydrate storage form used by animals
糖原: 動物使用的碳水化合物儲存形式
- All 3 examples are formed by combination of many **glucose** molecules!
3 個例子都是由多個**葡萄糖**分子組合而成的!
- Human digestive system cannot break down fibers! 人體消化系統無法分解纖維!

由單糖連接而成的長鏈(聚合物)



Major function of carbohydrate in human body:

碳水化合物在人體中的主要功能：

Major function in human cell is **providing a ready and easily used source of fuel.**

Glucose is usually the preferred fuel. When glucose is broken down in cell, chemical energy is released for other reactions in cell.

Excess glucose molecules are changed to glycogen or fat and stored.

在人體細胞內的主要功能是**提供現成且易於使用的燃料來源。**

葡萄糖通常是首選燃料，當葡萄糖在細胞中分解時，化學能被釋放出來用於細胞中的其他反應，過量的葡萄糖分子被轉化為糖原或脂肪並儲存起來。

Important Chemicals in Human Body

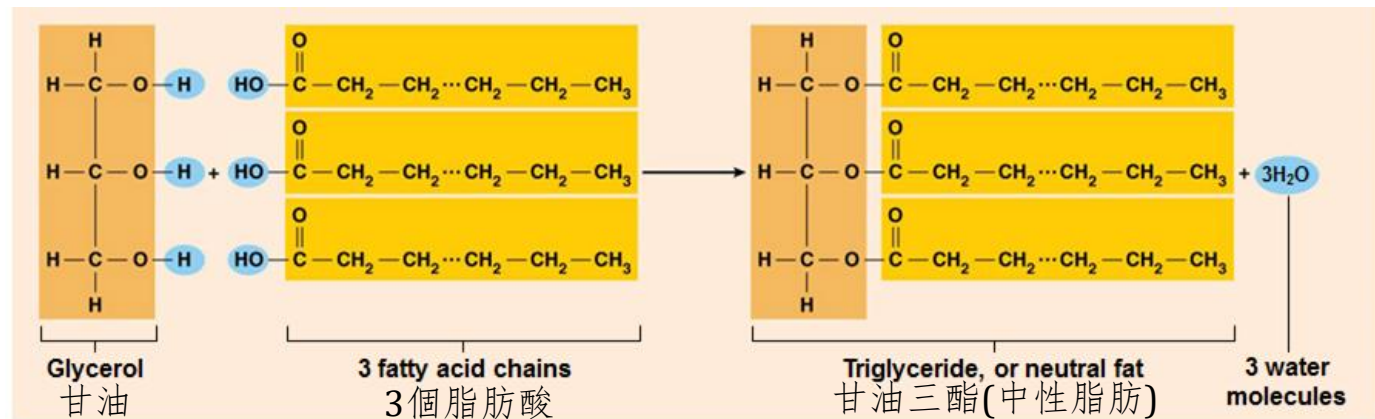
人體重要化學物質

- Lipids: 脂肪:
- Contain atoms C, H, O, and sometimes contain atom P
包含原子 C、H、O，有時包含原子 P
- Insoluble in water
不溶於水
- Main types: 主要類型:
 - **Triglycerides** (neutral fats) 甘油三酯(中性脂肪)
 - **Phospholipids** 磷脂
 - **Steroids** 類固醇

Important Chemicals in Human Body

人體重要化學物質

- **Triglycerides** (neutral fats) 甘油三酯(中性脂肪)
- 3 fatty acids + 1 glycerol molecule
3個脂肪酸+ 1個甘油分子

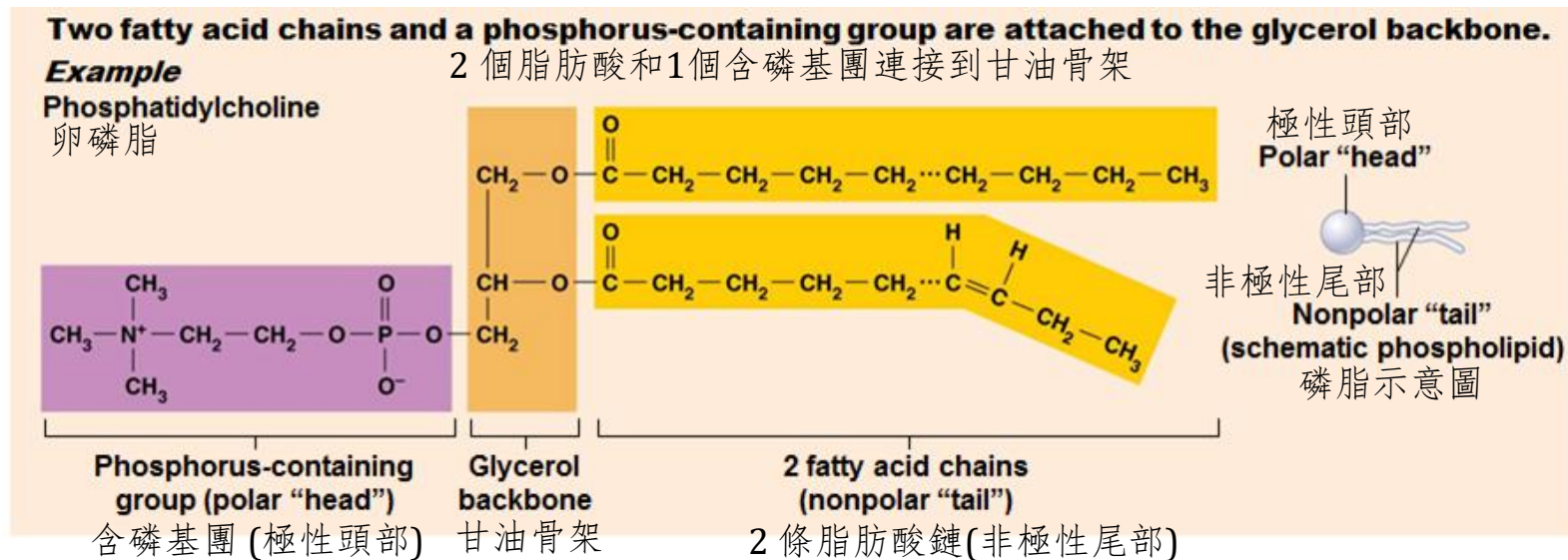


- Main functions 主要功能
- **Energy storage** (Triglyceride); **Fuel** (Fatty acid) 儲能(甘油三酯); 燃料(脂肪酸)
- **Insulation**: Subcutaneous fat reduce heat loss 絕緣: 皮下脂肪減少熱量流失
- **Protection**: Fat tissues surrounding organs to absorb shock
保護: 器官周圍的脂肪組織吸收震動

Important Chemicals in Human Body

人體重要化學物質

- **Phospholipids 磷脂**
 - 1 glycerol + 2 fatty acids + 1 phosphate group
1個甘油 + 2個脂肪酸 + 1個磷酸鹽基團
 - Forming **cell membrane** of human cell 形成人體細胞的**細胞膜**



Important Chemicals in Human Body

人體重要化學物質

- **Steroids 類固醇**

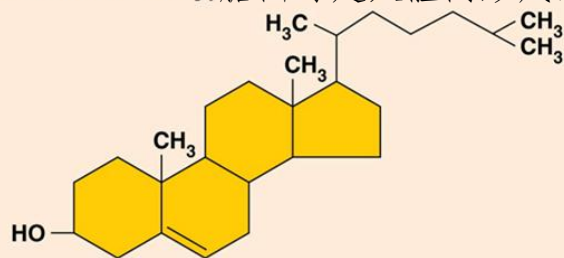
- Four interlocking ring structures 結構是由四個環狀的碳氫化合物互鎖而成
- Common steroids: 常見類固醇:
 - **Cholesterol**, Vitamin D, Steroid hormones, and Bile salts
膽固醇、維生素 D、類固醇激素和膽鹽
 - Cholesterol is used to make vitamin D, steroid hormones, and bile salts in human body
膽固醇用於在人體內製造維生素 D、類固醇激素和膽鹽

Four interlocking hydrocarbon rings form a steroid.

Example 四個環狀的碳氫化合物互鎖形成類固醇

Cholesterol (cholesterol is the basis for all steroids

formed in the body) 膽固醇是人體內形成的所有類固醇的基礎



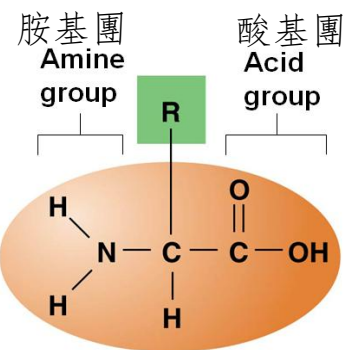
Important Chemicals in Human Body

人體重要化學物質

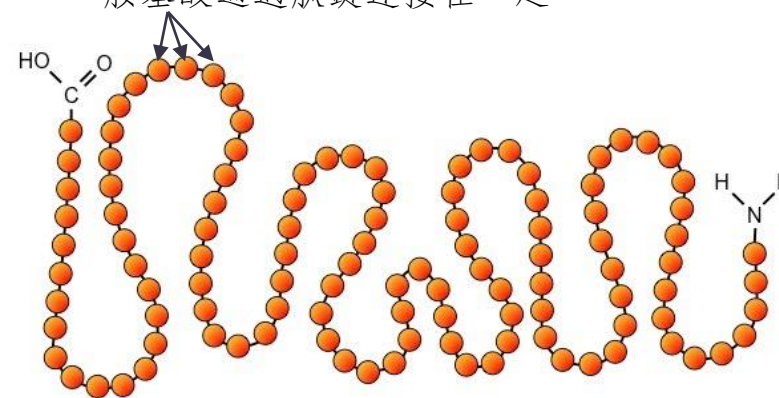
Typical structure
amino acid

- **Proteins: 蛋白質:**
- Comprise 20–30% of cell mass 佔細胞質量的 20-30%
- Contain atoms C, H, O, N, and sometimes atoms S and P 包含原子 C、H、O、N，有時包含原子 S 和 P
- Formed by **amino acids** held together by peptide bonds 由**胺基酸**以肽鍵結合在一起形成
- Four structural levels result in different shapes and functions 四個結構層次導致不同的形狀和功能

氨基酸的典型結構

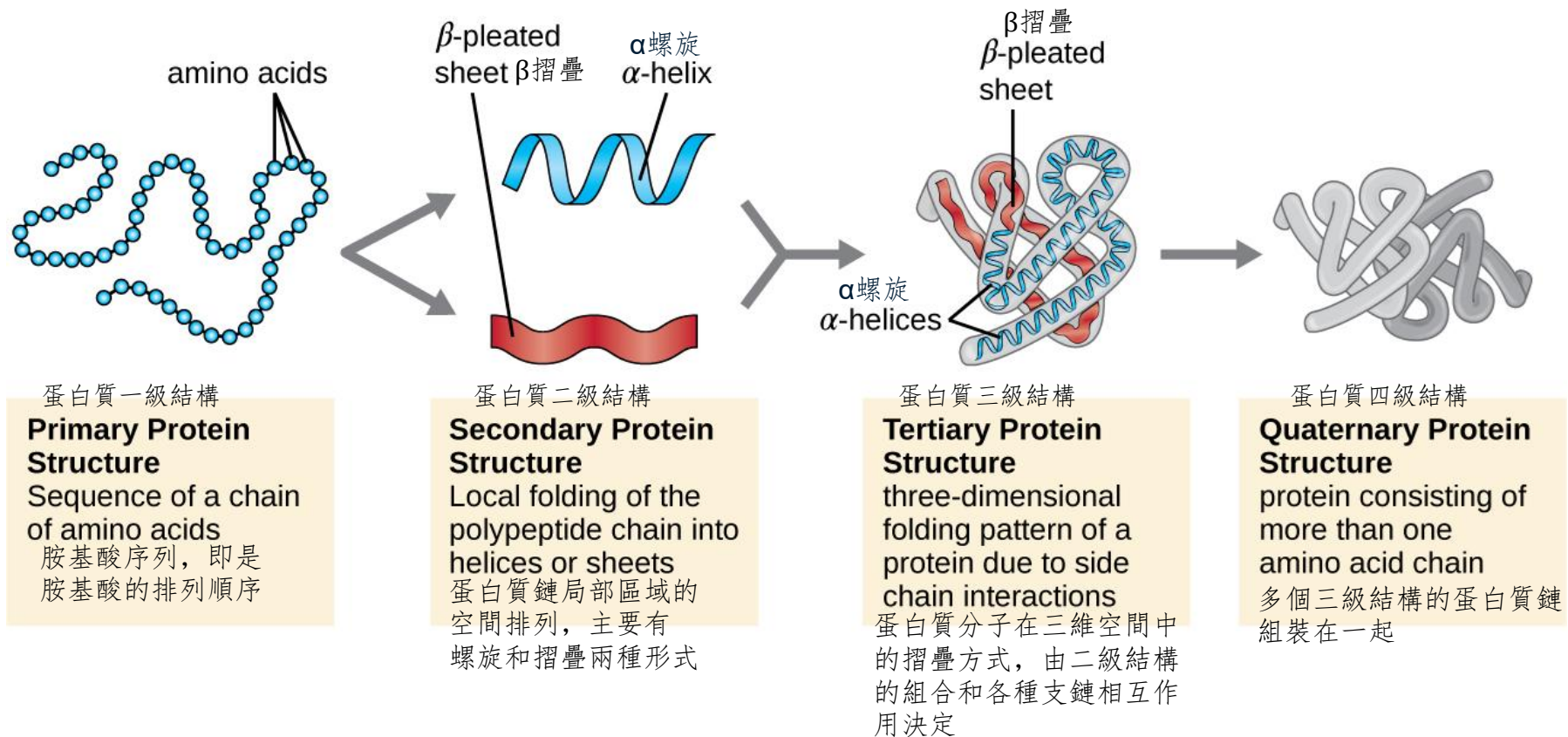


Amino acids held together by peptide bonds
胺基酸通過肽鍵連接在一起



Important Chemicals in Human Body

人體重要化學物質



The **3-dimensional folding shape** of a protein molecule is important for its function!

蛋白質分子的三維摺疊形狀對其功能很重要！

E.g. forming the specific active site of an enzyme! 例如，形成酶的特定活性位點！

Important Chemicals in Human Body

人體重要化學物質

- **Denaturation** of protein 蛋白質變性
 - Proteins unfold and **lose their functional 3-D shape**
蛋白質展開並失去其功能性三維形狀
 - Proteins become deactivated
蛋白質失活無效
 - Can be caused by 可能由以下原因引起
 - **extreme pH** (acid or alkaline)
極端 pH 值(酸或鹼)
 - increased temperature
溫度升高

Important Chemicals in Human Body

人體重要化學物質

- General Functions of Proteins: 蛋白質的一般功能:
- **Structural framework/ Mechanical support:** 結構框架/支撐:
E.g. **Collagens** in bones, cartilages, tendons, etc.
例如骨骼、軟骨、肌腱等中的**膠原蛋白**。
- **Movement: Contractile proteins** in muscle cells
活動: 肌肉細胞中的**收縮蛋白**
- **Enzymes:** Regulate rate of chemical reactions in human body
酶: 調節人體內化學反應的速率
- **Body defense: Antibodies** released by immune cells are used to defense against foreign substances
身體防禦: 免疫細胞釋放的**抗體**用於防禦外來物質
- **Membrane proteins:** forming cell membrane and acting as **receptor proteins** or **transporter proteins**
膜蛋白: 形成細胞膜並充當**受體蛋白**或**轉運蛋白**

Other Important Chemicals in Human Body

人體其他重要化學物質

- **Nucleic acids 核酸**
 - composed of C, H, O, N, and P 由 C、H、O、N 和 P 組成
 - the largest molecules in the body 體內最大的分子
 - Two major classes: 兩大類:
 - **Deoxyribonucleic acid (DNA) 去氧核糖核酸**
 - **Ribonucleic acid (RNA) 核糖核酸**

	DNA 去氧核糖核酸	RNA 核糖核酸
Major Function 主要功能	Hold genetic information 存放遺傳信息	Copy and carry information from DNA for protein synthesis 複製及攜帶DNA資訊用於蛋白質合成
Location in Cell 在細胞中之位置	Find almost entirely in nucleus 幾乎全部在細胞核中	Form in nucleus 在細胞核中形成 Find throughout cytoplasm 存在於整個細胞質中

L1 Revision Exercise L1 複習練習

- Screen the QR code below to complete all questions in this revision exercise.
掃描下面的二維碼以完成此複習練習中的所有問題。



References

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